

Syllabus

MAT 4233 — Modern Abstract Algebra Fall 2026 · UT San Antonio

Texts and Materials

Primary text

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*
- **Beardon** — Beardon, *Algebra and Geometry*. Ch. 3 (complex numbers, polar form, roots of unity) and Ch. 13 (Möbius transformations).

Supplementary reading

- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*. Not held by UTSA. Ch. 3 distributed as a handout; a copy is on reserve.

Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*. Survey reading for the final lecture: statements and examples; proofs optional.
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*. Notices of the AMS 55 (2008), no. 10. Co-reading for the Möbius lecture.

Course Information

Catalog Entry

MAT 4233 Modern Abstract Algebra

Basic properties and examples of semigroups, monoids, and groups, detailed study of permutation, dihedral, and congruence groups, cyclic groups, normal subgroups, quotient groups, homomorphism, isomorphism theorems, direct products of groups, rings and fields and their basic properties, ideals, polynomial rings.

Prerequisites

MAT 3233 Modern Algebra or equivalent

Credit Hours

3

Course Modality

Traditional in-person

Class Meetings

Duration	Wed 19 Aug – Thu 3 Dec
Campus	Main Campus
Location	MS 2.02.12
Time	Tue Thu 2:30 PM–3:45 PM

Learning Objectives

By the end of this course you should be able to:

- Define the notions of group, cyclic group, subgroup, normal subgroup, homomorphism, isomorphism.
- Define symmetric and permutation groups; determine the cycle decomposition and parity of finite permutations.
- Define the notions of ring, subring, domain, field, ideal, homomorphism, isomorphism.
- Define rings of polynomials over a field, the notion of degree, and irreducible factors.
- Explain the abstract algebraic properties of various rings and number systems including $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, modular rings $\mathbb{Z}/n\mathbb{Z}$, finite fields $\mathbb{F}_p$ and polynomial rings $\mathbb{F}[x]$.
- Explain the common and specific aspects of the division and Euclidean algorithms in the rings $\mathbb{Z}$ of integers, $\mathbb{Z}[i]$ of Gaussian integers, and polynomial rings $\mathbb{F}[x]$.
- Show that every Euclidean domain is a Unique Factorization Domain.
- Use properties of Gaussian primes and factorization in $\mathbb{Z}[i]$ to characterize sums of pairs of perfect squares.
- Communicate concepts in writing and debate technical arguments orally.
- Use a computer algebra system to for numerical and visual explorations in algebraic structures (e.g., the RSA cryptosystem and geometric transformation groups).

About the Instructor

Eduardo Dueñez, Associate Professor of Mathematics.

Department	Mathematics
Office	FLN 4.01.11
Student hours	TBD
Email	eduardo.duenhez@utsa.edu
Homepage	https://supernumero.us/about

Email is the preferred method of communication.

Assessment and Grading

Weight of course activities

Component	Weight	
Q&A sessions	40%	5 sessions, 8% each
Problem sets	15%	5 sets, 3% each
Midterm Exam 1	20%	Thu 1 Oct in class
Midterm Exam 2	25%	Thu 3 Dec in class

Q&A sessions

Five class meetings throughout the semester (roughly every other Thursday except when adjacent to an exam) will be *Q&A sessions* in which each student will be asked to explain and defend their solution to a homework problem submitted as part of a written assignment. You will only be asked questions related to an assigned problem and about the solution you submitted.

Each Q&A session is scored 0–3:

- **3** Correct explanation to a correct solution, demonstrating full understanding of the concepts and all aspects of the solution submitted.
- **2** A completely correct solution submitted is explained to an incomplete degree; **Or**: an unfinished solution, with a thorough and articulate account of what you tried, explaining where and why you got stuck while demonstrating knowledge of the background and reading necessary to solve the problem.
- **1** Correct solution inadequately explained revealing little or no understanding; *or*: partial solution accompanied by an incomplete but partly correct oral explanation.
- **0** Nothing submitted; *or*: work submitted (even if fully correct) is backed by no cogent explanation nor knowledge of concepts involved.

A correct answer you cannot explain is worth less than an explainable honest failure.

Note: A Q&A average ≤ 1.0 automatically caps the course grade to the level of ‘D’ (at most).

Problem sets

Five sets, each with three or four theoretical problems plus one SageMath computer exploration. **Maximum three pages.** Do *not* submit code; include the relevant snippets and output as part of your written submission. (SageMath problems are also included in Q&A.)

Each item will be scored *only* based on the **degree of completion** (not on correctness): **2** for a nominally complete solution, **1** for a clearly partial one, **0** for nothing submitted or for obvious filler with no substance.

Note: Any submission containing non-human-readable content, stray characters, or artifacts revealing non-proofread AI output scores 0 for that item.

Use of Artificial Intelligence (AI)

AI assistance is allowed in preparing work for submission, as are any solution keys, collaboration, and internet sources. You are ultimately responsible for your submissions; any part of your work that has clearly not been proofread (e.g., includes non-human readable characters or LLM artifacts) automatically earns a zero grade. *Include one line noting what resources you used in your submission.*

This policy is by design, and it reflects how the course is graded: 85% of the grade comes from Q&A sessions and in-class exams where only what is truly understood and can be explained counts. Assignments submitted but not written/understood by you result in failing grades in Q&A sessions and exams. Use AI to learn faster —not to prepare and turn in papers you cannot defend, blocking you from passing exams as well.

All audiovisual (or audio only, or video only) recordings of lectures are explicitly forbidden; all uses of AI agents or apps to capture, process, or analyze any lecture contents are **strictly forbidden**. The only *limited* exceptions are *explicitly* allowed recordings per official memorandum of the office of Student Disability Services. Under no circumstances is any processing or analyzing of such recordings allowed. Any sharing of confidential course materials with individuals who are not enrolled students in the course is a violation of [FERPA privacy laws](#).

Exams

Both exams are traditional in-class (written and closed book). Exams are based **primarily on the assigned problem sets** —expect some context and phrasing to differ from the homework, plus a minority of other unseen problems. Memorizing solutions will not result in good exam grades; preparing by understanding the assigned problems will make the exams straightforward and earn high Q&A grades as well.

Attendance and missed work

No extensions on problem sets. A *single* missed Q&A session in the semester may be made up during office hours within seven days (no documentation is required). *No Q&A scores are dropped.*

Final grade ranges

Grade	Range	Grade	Range
A	[90%, 100%]	A-	[85%, 90%]
B+	[80%, 85%]	B	[76%, 80%]
B-	[72%, 76%]	C+	[69%, 72%]
C	[65%, 69%]	C-	[60%, 65%]
D	[50%, 60%]	F	[0%, 50%]

Time Commitment Expectations

Expect an average of about **9 hours per week**, and no fewer than 7.

- Attend class meetings (**3 hours per week**)
- Reading the assigned textbook sections (**2–3 hours per week**)
- Problem sets and the Sage exploration (**2–3 hours per week**)
- Office hours and study group (**1 hours per week**)

Additional Course Information

Each student should be intimately familiar with the contents of any work submitted for grading. The instructor has the right to request adequate **verbal explanation** of methodology and content for any submitted work. Credit will not be awarded when such an explanation is requested but not satisfactorily provided.

Course Schedule

Date	Activity	Topic
Thu 20 Aug		Course launch; divisibility and the division algorithm
Tue 25 Aug		Primes, the Fundamental Theorem of Arithmetic, gcd and lcm
Thu 27 Aug		Semigroups, monoids, groups — all of them $\mathbb{Z}/n\mathbb{Z}$
Tue 1 Sep		Subgroups; the unit group $(\mathbb{Z}/n\mathbb{Z})^*$
Thu 3 Sep	Q&A	Defense 1
Tue 8 Sep		Group homomorphisms; the order of an element
Thu 10 Sep		Cyclic groups; primitive roots
Tue 15 Sep		Cosets and Lagrange's theorem
Thu 17 Sep	Q&A	Defense 2
Tue 22 Sep		Normal subgroups and quotient groups
Thu 24 Sep		The homomorphism theorem
Tue 29 Sep		Euler's theorem, fast modular exponentiation, and RSA end to end
Thu 1 Oct	Exam	Midterm Exam 1 — Segment 1
Tue 6 Oct		Rings and subrings

Date	Activity	Topic
Thu 8 Oct		Ring homomorphisms
Thu 15 Oct		Ideals
Tue 20 Oct		Quotient rings
Thu 22 Oct	Q&A	Defense 3
Tue 27 Oct		Fields and skew fields; F_p
Thu 29 Oct		Integral domains and the field of fractions
Tue 3 Nov		Unique factorization: UFDs
Thu 5 Nov	Q&A	Defense 4
Tue 10 Nov		PIDs and Euclidean domains
Thu 12 Nov		Complex numbers: arithmetic, the plane, polar form, roots of unity
Tue 17 Nov		Mobius transformations: the formula, composition as matrix multiplication, circles to circles, the Riemann sphere, fixed points
Thu 19 Nov	Q&A	Defense 5
Tue 24 Nov		The Gaussian integers: norm, division with remainder, irreducibles, and sums of two squares — a survey
Tue 1 Dec	<i>Review</i>	Review — the exam pool
Thu 3 Dec	Exam	Midterm Exam 2 — Segment 2 weighted

Readings for each meeting and assignment due-dates are on the [Weekly Schedule](#) (also available as a [printable PDF](#)).

Department and University Policies

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu.

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law.

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes or corrections to the course materials, assignment dates, or other updates will be communicated to students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information](#) webpage for resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success

coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.

- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support page](#).
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#), or call (210) 458-4157.
- Your well-being is important. Support, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site.
- [Student Assistance Services](#) offers confidential support and personalized guidance.
- The [Roadrunner Pantry](#) provides free access to groceries, toiletries, and other essentials.
- Students are responsible for ensuring their work is consistent with UTSA's standards for academic integrity. Review [Section 203 of the Student Code of Conduct](#).
- Visit [UT San Antonio Libraries and Museums](#) for journals, research tutorials, and your department's librarian.